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SamsaRaLight: An R package to estimate tree light interception using ray-tracing

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Summary

The *{SamsaRaLight}* R package (Figure 1) provides a simplified and efficient R interface to the SamsaraLight ray-tracing model (Benoit Courbaud et al., 2003), enabling explicit simulation of light competition in forest stands. It allows users to estimate the energy absorbed by individual trees, the light transmitted to the forest floor, and optionally the radiation reaching user-defined locations (virtual sensors). Forest stands are represented in 3D, with individual crowns modelled using symmetric or asymmetric geometric shapes and stand attributes including size, slope, and aspect. Direct and diffuse radiation is discretised according to stand location and cast through the canopy, allowing light attenuation across successive crowns to be explicitly simulated. Outputs include light interception, transmission, and competition variables at tree, ground, and sensor levels, supporting studies of tree growth, mortality, and regeneration dynamics within reproducible and optimised R workflows.

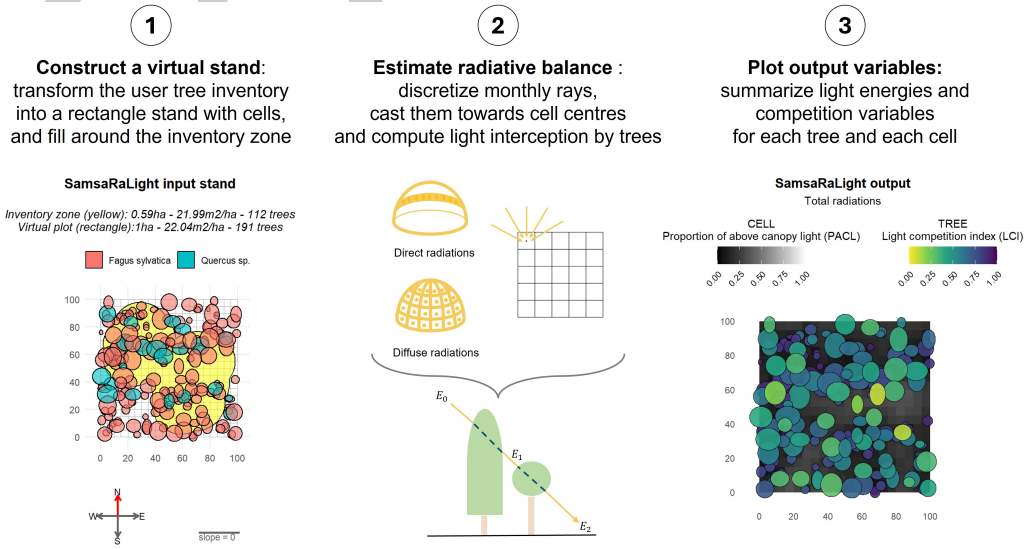

Figure 1: Global aim of the *{SamsaRaLight}* R package.

Statement of need

Light is a key resource driving tree growth and mortality (Beauchamp et al., 2025), consequently affecting forest stand dynamics (Binkley et al., 2013) and canopy structure (Touzot et al.,

2025). Because light penetrates forest canopies primarily from above in a predominantly vertical direction, larger trees intercept and attenuate incoming radiation first (Monsi et al., 1973). Therefore, competition for light is called asymmetric: the largest individuals monopolise a disproportionate share of resources relative to their size, thereby strongly reducing access to light for smaller individuals (Schwinning & Weiner, 1998; Weiner, 1990). Numerous forest radiative transfer models have therefore been developed to represent canopy structure and light absorption processes (Ligot, Balandier, Courbaud, & Claessens, 2014). Accurately representing this process can be carried out with ray-tracing approaches that explicitly account for light path length across successive tree crowns.

The SamsaraLight ray-tracing model, developed by Courbaud et al. (2003), is a widely used models for this purpose in forest ecology. It has been applied to study the effects of light interception on tree growth and mortality dynamics (Beauchamp et al., 2025), as well as to quantify light transmittance to the forest floor and its influence on regeneration processes (Ligot, Balandier, Courbaud, Jonard, et al., 2014; Ligot et al., 2016). Explicitly coupling light competition with forest simulators is essential to study forest stand evolution over time and how silvicultural strategies influence stand structure, dynamics, and regeneration Barrere et al. (2024). For this reason, SamsaraLight has been integrated as a core module in numerous forest simulators including Samsara2 (B. Courbaud et al., 2015), PDG-ARENA (Rouet et al., 2025), HETEROFOR 1.0 (Jonard et al., 2020; Wergifosse et al., 2020), or RReShar (Barrere et al., 2024).

State of the field

The SamsaraLight ray-tracing model was originally developed within the CAPSIS platform (Dufour-Kowalski et al., 2012), a Java-based framework simplifying the development of forest simulators. However, the tight coupling of SamsaraLight to CAPSIS limits its independent use and integration into common statistical workflows. To address this, Fortin (2020) introduced the R packages *{J4R}* and *{RCapsis}*, enabling CAPSIS models to be accessed from R. Despite this advance, the approach remains technically demanding, requiring interaction with external software, intermediate files, and multi-language pipelines, which restricts accessibility for many users.

Many researchers working on forest dynamics and regeneration increasingly rely on R for model fitting, simulation analysis, and reproducible workflows. While some forest models have been fully reimplemented in R (*e.g.*, SurEau-Ecos (Ruffault et al., 2022)), ray-tracing approaches remain computationally intensive and difficult to use efficiently without high-performance and accessible implementations. By combining a user-friendly R interface with an efficient C++ back-end, the *{SamsaRaLight}* R package addresses these limitations, simplifying the explicit representation of light interception and transmission and enabling direct use in studies of tree growth, mortality, and regeneration.

Software design

Native R workflow with pre- and post-processing tools

The *{SamsaRaLight}* R package provides a fully native R implementation of the SamsaraLight ray-tracing model. Forest stands are defined using standard R formats, with dedicated functions to check data and construct virtual stands from tree inventories. Based on stand geographic coordinates, the package facilitates access to input radiative data by retrieving monthly global radiation and diffuse-to-global ratios from the European PVGIS database (Huld et al., 2012). Simulations are executed with default settings suitable for most applications, while allowing advanced users to adjust ray-tracing parameters and internal model behaviour. Standard R methods for S3 objects (`print()`, `summary()`, and `plot()`) enable inspection and visualisation of both input stands and output light variables, with graphical outputs generated using *{ggplot2}* [(**?**)

High computational performance through C++ implementation and parallelisation

To ensure efficient computation, the core of the SamsaraLight model is implemented in C++ and interfaced with R using *{Rcpp}* (Eddelbuettel et al., 2025). This design leverages the computational performance and memory management of C++ while preserving the flexibility of R-based workflows. Ray-tracing calculations can be executed in parallel using shared-memory parallelisation with OpenMP (Dagum & Menon, 1998), with fine control over the number of cores and the ability to activate or deactivate parallel execution depending on computational constraints.

Documentation and learning resources

The *{SamsaRaLight}* package is accompanied by an online documentation website available at https://natheob.github.io/SamsaRaLight/, generated with *{pkgdown}* (Wickham et al., 2025). The documentation provides a detailed description of the underlying model and a set of progressive vignettes designed to support learning and practical use. These tutorials are based on real forest datasets distributed with the package and guide users through typical workflows.

Research impact statement

The *{SamsaRaLight}* R package has multiple utilities for researchers, students, and field foresters, enabling broader and more user-friendly access to ray-tracing models. Its applications include: (1) simplifying workflows for forest light competition studies; (2) serving as a teaching tool through progressive tutorials that guide students in understanding and applying ray-tracing models; and (3) providing accessible visualization tools for estimating light distribution in forest stands directly from tree inventories, with future plans to extend usability via a Shiny application for non-programmer users (Chang et al., 2025).

The *{SamsaRaLight}* R package could facilitate links with emerging technologies and research fields. In particular, the increasing availability of LiDAR data in forest ecology (Lines et al., 2022) creates opportunities to couple ray-tracing light models with LiDAR-based crown reconstruction tools such as ITSMe (Terryn et al., 2023) and LidaRtRee (Monnet, 2025). More broadly, providing an accessible R implementation may foster applications beyond forestry, including agroforestry, where the SamsaraLight model has already been used for estimating tree shading influences on crop production (Dupraz et al., 2019).

AI usage disclosure

Generative AI tools were used in a limited support role. The core C++ implementation of the SamsaraLight model and the R interface were fully developed by the authors before the widespread availability and use of generative AI tools. ChatGPT was later used to help refine function documentation, tests, and user-facing interfaces for inspection. The initial manuscript and online documentation were first written by the authors without AI assistance, and ChatGPT and DeepL Write were subsequently used only to improve clarity, conciseness, and reduce redundancy in the text. All AI-assisted content was reviewed, edited, and validated by the authors, who take full responsibility for the software and the paper.

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